

Simplicial approximation with Delaunay complexes

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Classical subdivisions

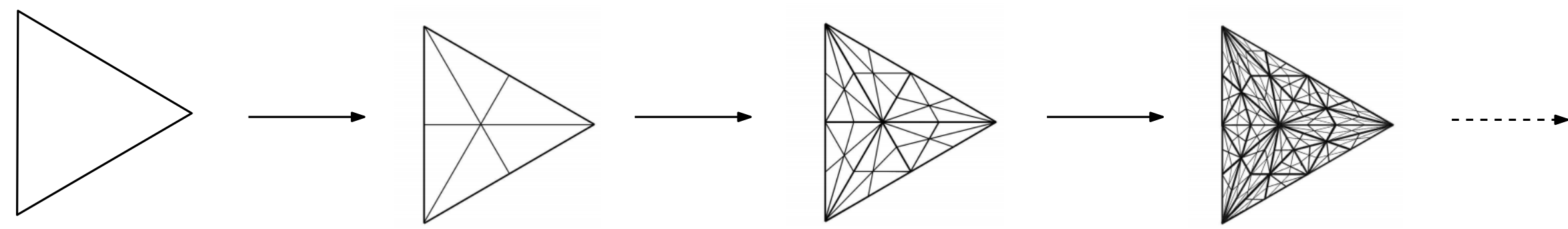
Barycentric subdivision

Decomposes σ^d into a simplicial complex with $2^{d+1} - 1$ vertices and $(d+1)!$ facets.

New vertices: the equal-weight barycenters of the faces. That is, for $\sigma = \{v_0, \dots, v_d\}$,

$$V(\text{sub}(\sigma)) = \left\{ \frac{1}{k} \sum_{\ell=1}^k v_{i_\ell} \mid i_1, \dots, i_k \in \{0, \dots, d\} \text{ distinct, } 1 \leq k \leq d+1 \right\}.$$

Contraction property: $\max \text{diam}(\text{sub}(\sigma)) \leq \frac{d}{d+1} \text{diam}(\sigma).$



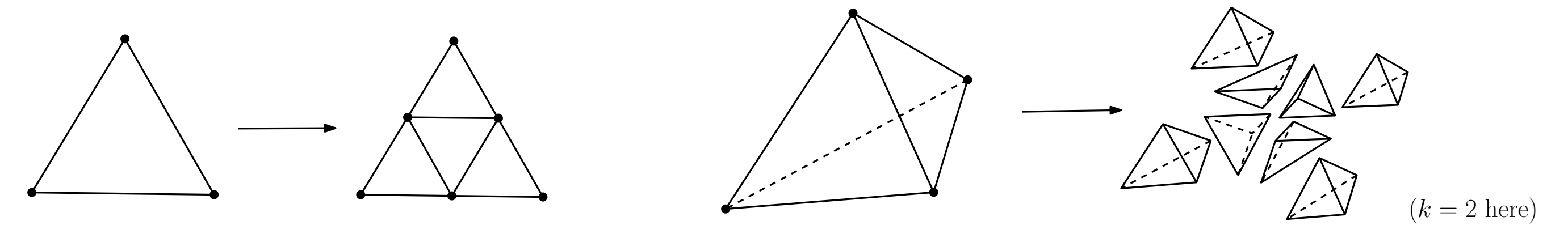
k -Edgewise subdivision

Decomposes σ^d into a simplicial complex with $\binom{d+k}{k}$ vertices and k^d facets.

New vertices: the k -fold barycenters of the vertices:

$$V(\text{sub}(\sigma)) = \left\{ \frac{1}{k} \sum_{\ell=1}^k v_{i_\ell} \mid i_1, \dots, i_k \in \{0, \dots, d\} \text{ repetitions allowed} \right\}.$$

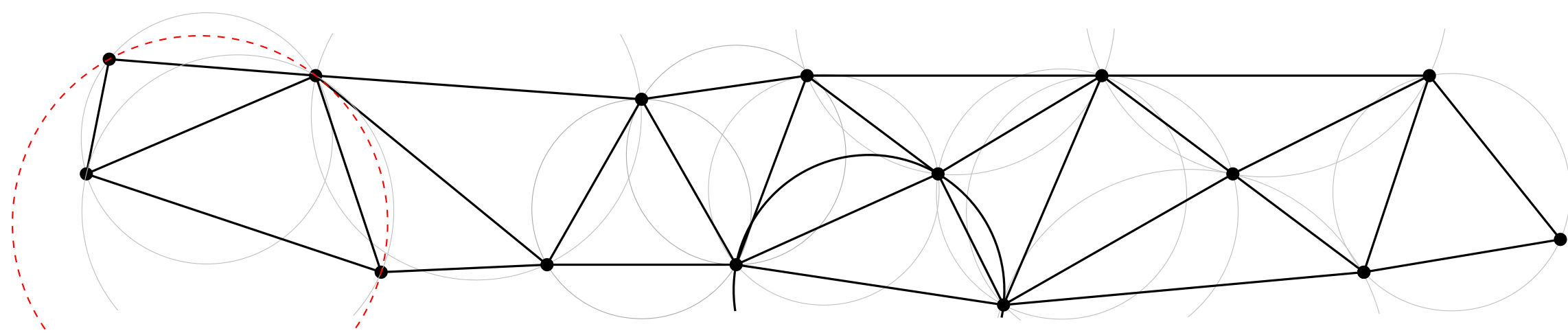
Contraction property: $\max \text{diam}(\text{sub}(\sigma)) \leq \min \left\{ 1, \frac{\lfloor \frac{d+1}{2} \rfloor}{k} \right\} \text{diam}(\sigma).$



Delaunay refinement

Delaunay complexes

Let $X \subset \mathbb{R}^d$ be finite. $\text{Del}(X)$ is the simplicial complex whose facets are the subsets of $d+1$ points whose circumscribing open ball is empty of points of X .



Spherical Delaunay complexes

Let $X \subset S^d$ be finite. $\text{Del}(X)$ is the simplicial complex whose facets are the subsets of $d+1$ points whose **spherical** circumscribing open ball is empty of points of X .

Lemma: $\text{Del}(X)$ coincides with the boundary of the convex hull of X in $\mathbb{R}^{d+1}$.

Lemma: $\max \text{diam}(\text{Del}(X)) \leq 2 \text{cov rad}(X)$, where the **covering radius** is

$$\text{cov rad}(X) = \sup_{y \in S^d} \inf_{x \in X} d(x, y).$$

Delaunay refinement

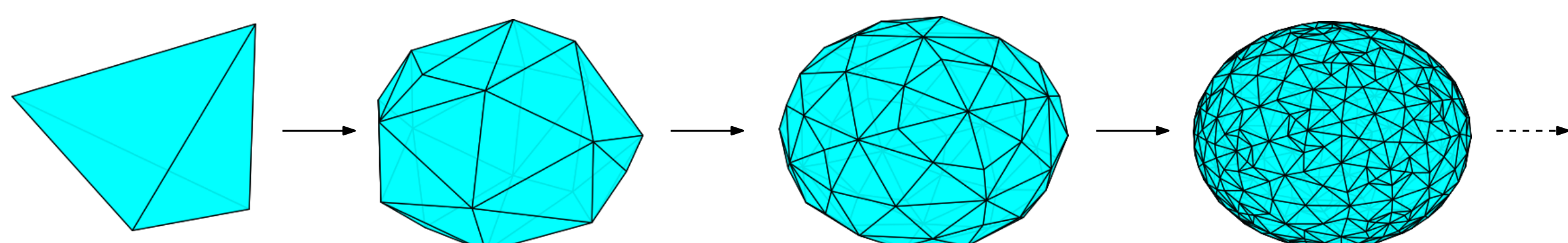
Given an initial $\text{Del}(X_0)$, with $X_0 \subset S^d$, one obtains a new triangulation by choosing **Steiner points** $Y_0 \subset S^d$ and building the Delaunay complex of $X_1 = X_0 \cup Y_0$.

This construction can be iterated, yielding a sequence of Steiner points

$$Y_0, Y_1, Y_2, \dots$$

and a sequence of spherical Delaunay complexes

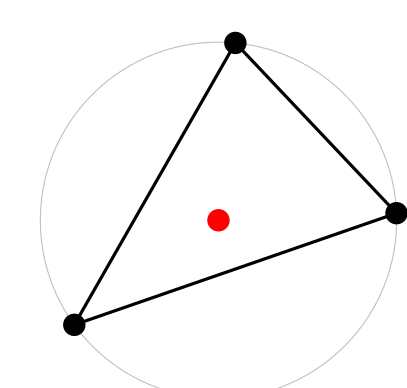
$$\text{Del}(X_0), \text{Del}(X_1), \text{Del}(X_2), \dots$$



Steiner points

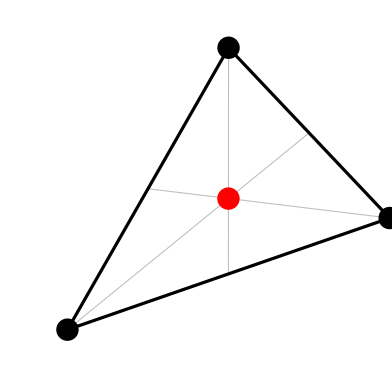
Minicenter

Add the minicenter of each facet



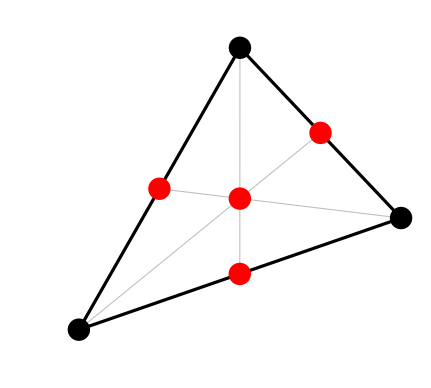
Centroid

Add the barycenter of each facet



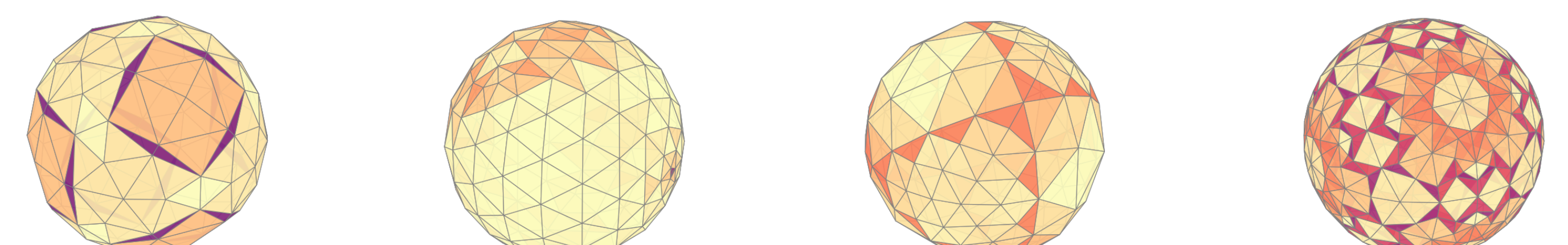
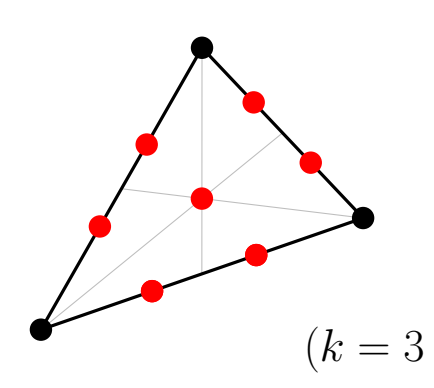
Barycentric

Add the barycenter of each simplex



k -Edgewise

Add all k -fold barycenters of each facet



Theorem

Let the initial sample $X_0 \subset S^d$ be sufficiently dense so that $\text{cov rad}(X_0) \leq \pi/3$. Then the n -th iteration of Delaunay refinement satisfies

$$\text{cov rad}(X_n) \leq \tilde{\alpha}^n \text{cov rad}(X_0), \quad \tilde{\alpha} = \frac{\alpha}{\cos(\text{cov rad}(X_0))},$$

where α depends on the chosen refinement.

Method	Minicenter	Centroid	Barycentric	k -Edgewise
α	$\frac{1}{\sqrt{2}}$	$\begin{cases} \frac{2}{3} & \text{if } d=2, \\ \frac{d^2-1}{d^2+1} & \text{if } d \geq 3 \end{cases}$	$\frac{1}{2}$	$\sqrt{\frac{s(d+1-s)}{dk^2}},$ $s = \min \left\{ k, \left\lfloor \frac{d+1}{2} \right\rfloor \right\}$

Approximate Carathéodory Theorem

Classical formulation

Let $X = \{x_1, \dots, x_n\} \subset B^d$ and $y \in \text{conv}(X)$. For every $k \geq 1$, there exists a k -tuple $(x_{i_1}, \dots, x_{i_k})$ (repetitions allowed) such that

$$\left\| y - \frac{1}{k} \sum_{\ell=1}^k x_{i_\ell} \right\| \leq \frac{1}{\sqrt{k}}.$$

Improvement

The optimal constant is

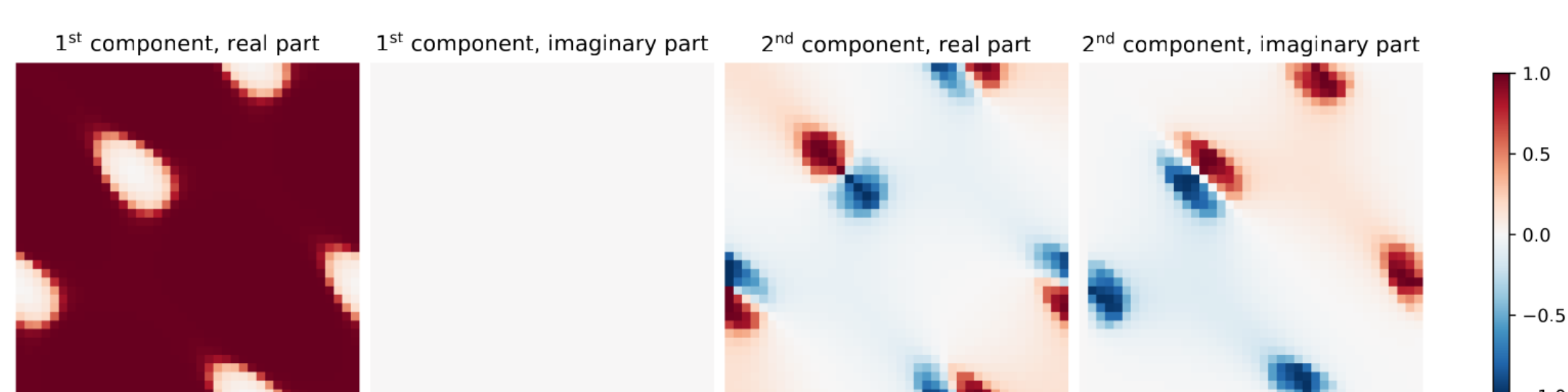
$$\zeta(d, k) = \sqrt{\frac{s(d+1-s)}{dk^2}}, \quad s = \min \left(k, \left\lfloor \frac{d+1}{2} \right\rfloor \right).$$

Simplicial approximation

Characteristic classes

Haldane Bundles dataset: complex line bundles on the torus T^2 [Tipton et al., 2024].

They are subbundles of $X \times \mathbb{C}^2 \rightarrow X$. We are given a section $T^2 \rightarrow \mathbb{C}^2$ (with zeros).

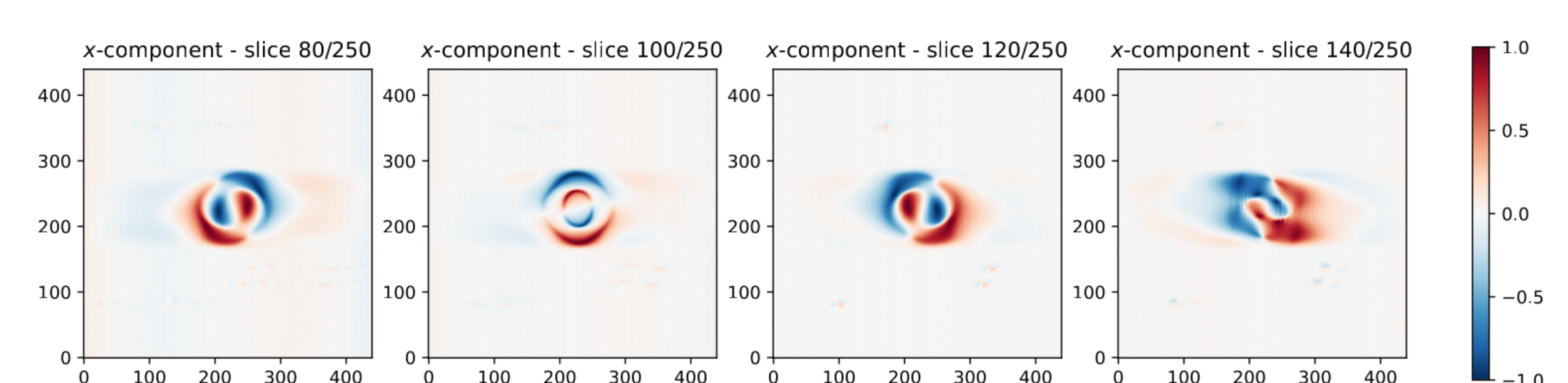


Projectivize to get a classifying map $f: T^2 \rightarrow \mathbb{CP}^1$.

The first **Chern class** can be computed via $f^*: H^2(T^2; \mathbb{Z}) \leftarrow H^2(\mathbb{CP}^1; \mathbb{Z})$.

Hopfion

Data from the MARTApp software [Herguedas-Alonso, 2025]: a unit vector field $\mathbb{R}^3 \rightarrow S^2$.



Compactify to get a map $f: S^3 \rightarrow S^2$. We build the mapping cone $C(f)$ and compute:

$$H^0 \cong \mathbb{Z}, \quad H^1 = 0, \quad H^2 \cong \mathbb{Z}, \quad H^3 = 0, \quad H^4 \cong \mathbb{Z}.$$

Moreover, $\alpha_2 \smile \alpha_2 = \alpha_4$ (computation using HAP). Hence f has **Hopf invariant 1**.